Joules per Correct Answer: Fine-Tuning, Quantization, Batching and Speculative Decoding for Small Code Models

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Abstract

Small code models are attractive because they run where the code is: on a developer's laptop, without a GPU. Their efficiency is usually reported as tokens per second, which ignores whether the answers are right and how long they are. We measure *energy per correct answer* for a 1.5B code model (Qwen2.5-Coder-1.5B-Instruct and a QLoRA fine-tune of it) on code generation and bug repair, on a laptop CPU (RAPL) and on an NVIDIA T4 (board power), and decompose it into four levers: fine-tuning, weight quantization, request batching and speculative decoding. On the CPU the levers compound: from the base model in F16 to the fine-tune in Q8 with n-gram lookup, energy per solved HumanEval+ problem falls from 962 J to 288 J with no significant change in accuracy (106 vs. 102 of 163 solved, $p = 0.63$), and to 189 J with Q4 weights (99 solved, $p = 0.35$); fine-tuning alone cuts it by a third, by making answers shorter. Speculative decoding behaves differently on the two devices: a 0.5B draft model is accepted 79–91% of the time on the CPU yet saves at most 6% energy, while draft-free n-gram lookup saves 21–25% on the CPU and 50–67% on the T4. Fine-tuning also moves the target away from its stock draft: acceptance is lower for the fine-tune than for the base model at every draft length, and with 8 drafted tokens the fine-tune repairs bugs more slowly than without speculation while the base model still gains. On the GPU, serving 8 requests at a time lowers energy per correct answer 6–9×, more than any decoding change. Finally, what fine-tuning teaches depends on its bugs: trained on one-line mutations the model learns to patch one token; with test-verified realistic bugs on a 3B base, benchmark repair rises from 50% to 63% and the tests the model writes catch the bug in 82% instead of 60% of problems. A check we registered before running it qualifies both gains: on 382 real bugs from 2024 programming contests the repair gain is +0.8 points ($p = 0.79$) and is not confirmed, and the test-writing gain holds with more output room but comes mostly from finishing the file. All answers, logs and scripts are public, and every accuracy comparison is a paired exact test on the same problems.

1 Introduction

A code assistant that runs locally keeps source code on the developer's machine and costs nothing per query, but it has to live within a laptop's CPU, memory and battery. For that setting the useful question is not how fast a model emits tokens but how much energy it spends per *correct* answer: a model that writes twice as many tokens, or gets fewer problems right, is more expensive even at the same speed.

We study one model family under that metric and ask which deployment choices move it, and by how much. We take Qwen2.5-Coder-1.5B-Instruct [3] and a QLoRA [2] fine-tune of it trained for test writing, repair and review, and measure code generation (HumanEval+, MBPP+ [7]) and bug repair (HumanEvalFix [9]). Our contributions:

- A decomposition of energy per correct answer into four levers, fine-tuning, quantization, batching and speculative decoding, measured on the same benchmarks on a laptop CPU and a T4 (Sections 3–6).
- Evidence that speculative decoding's value depends on the device more than on draft acceptance: a well-accepted 0.5B draft model barely helps on a CPU, where draft-free n-gram lookup [11] is the better choice; on a GPU both help, n-gram lookup most (Section 4). This is consistent with the observation that speculative speed-up is governed by draft latency rather than draft accuracy [13].
- Evidence that fine-tuning lowers the acceptance of a stock draft model, enough to make long drafts a net loss for the fine-tune (Section 5).
- A check that the fine-tuning gains replicate with a second seed (Section 7).
- Evidence that one-line mutation data teaches the shape of its bugs, and that realistic, test-verified bugs on a 3B base raise benchmark repair by 13–15 points (Section 8), with a pre-registered check showing that this gain does not carry over to real bugs written by people (Section 9).

2 Setup

Models. The base model is Qwen2.5-Coder-1.5B-Instruct (revision `2e1fd39`). The fine-tune adds a rank-16 QLoRA adapter on all projections, trained for 2 epochs on 14,151 execution-verified examples in four formats (repair of injected bugs, test writing, review of bug/clean pairs, execution traces) with 13-gram decontamination against all evaluation sets, on one T4 in 2 h. For deployment the adapter is merged and exported to GGUF in F16, Q8_0 and Q4_K_M. The draft model for speculative decoding is Qwen2.5-Coder-0.5B-Instruct, not fine-tuned.

Benchmarks. HumanEval+ (163 problems; HumanEval/32 is excluded because its reference test fails for any solution), MBPP+ (377) and HumanEvalFix in Python (164). All decoding is greedy with 768 new tokens; an answer is correct if it passes all tests, run in a sandbox without network.

Hardware and energy. The CPU is a laptop Intel Core i5-10210U (4 cores, 16 GB RAM), running llama.cpp (`987498f`) with 4 threads, one request at a time, prompt cache and memory mapping off. CPU energy is the RAPL package counter read per request. The GPU is an NVIDIA T4 (16 GB) on Kaggle, running vLLM 0.29 [4] in FP16 with CUDA graphs. GPU energy is the trapezoidal integral of `nvidia-smi` board power, sampled at 1 s, over each run. Both include idle draw: they are the energy the device spent while the requests were served.

Metric and tests. Energy per correct answer is the total energy of a run divided by the number of problems solved. Accuracy differences are tested with the exact McNemar test on the same problems; we call a difference significant at $p < 0.05$ and report counts.

3 Fine-tuning and quantization on the CPU

Table 1 shows two independent levers, and Figure 1 how they combine with speculative decoding. **Quantization** works as expected for a memory-bound CPU: decode speed follows model size (Q8

	base F16	base Q4	FT F16	FT Q8	FT Q4
HumanEval+ pass@1 (%)	65.0	67.5	64.4	63.2	60.1
MBPP+ pass@1 (%)	56.2	60.5	60.2	59.4	58.1
HumanEvalFix pass@1 (%)	–	22.0	32.3	–	29.9
Decode speed (tok/s)	5.8	16.8	5.8	10.4	16.1
Output tokens, HumanEval+	227	209	145	143	112
J per correct, HumanEval+	962	332	638	396	234
J per correct, MBPP+	495	169	238	146	107
J per correct, HumanEvalFix	–	756	833	–	471
Peak RAM (GB)	3.27	1.29	3.21	1.79	1.16

Table 1: Laptop CPU, plain decoding. FT: the merged QLoRA fine-tune. Q4 is Q4_K_M.

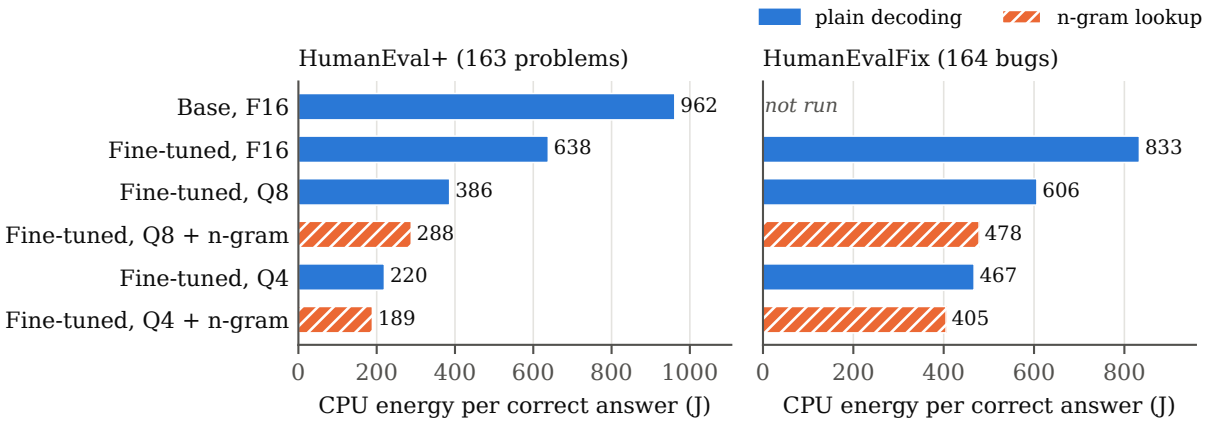


Figure 1: Energy per correct answer on the laptop CPU, from the base model in F16 to the fine-tune in Q4 with n-gram lookup. Base and fine-tuned F16 are from the quantization runs of 14 October, Q8 and Q4 from the speculative runs of 19 October (Section 4); the base model in F16 was not run on HumanEvalFix.

1.8×, Q4 2.8× faster than F16). Q8 is lossless against F16 (2 of 163 and 9 of 377 answers change, $p \geq 0.5$); Q4 costs the fine-tune 2–4 points on all three benchmarks, none significant alone.

Fine-tuning saves energy without making the model faster: the fine-tune writes about a third fewer tokens (145 vs. 227 on HumanEval+, 50 vs. 102 on MBPP+), because its training targets are bare code, while the base model explains its answer. At equal precision this is 34% less energy per solved HumanEval+ problem and 52% less per solved MBPP+ problem, at accuracy that is equal (HumanEval+, $p = 1$) or higher (MBPP+, +4.0 points, $p = 0.077$). On repair the fine-tune solves more problems with shorter answers: at Q4 it needs 471 J per fixed bug against 756 J for the base model, and fixes 8 points more ($p = 0.024$).

4 Speculative decoding

Table 2 compares a 0.5B draft model with n-gram lookup. On the **CPU** the draft model is accepted 79–91% of the time, yet speeds decoding up by at most 1.19× and saves at most 6% energy; on repair it costs 9–10% more than plain decoding. The draft runs on the same four cores as the target, scoring a block of drafted tokens is not free on a CPU, and the draft also has to read the prompt,

		HumanEval+		HumanEvalFix	
Device	Setting	Speed-up	J/correct	Speed-up	J/correct
CPU, FT Q8	none	1.00	386	1.00	606
	draft Q4, 4 tokens	1.17	367 (−5%)	1.10	665 (+10%)
	draft Q4, 8 tokens	1.19	362 (−6%)	1.09	663 (+9%)
	draft Q8, 8 tokens	1.10	378 (−2%)	1.03	658 (+9%)
	n-gram	**1.48**	**288 (−25%)**	**1.57**	**478 (−21%)**
CPU, FT Q4	none	1.00	220	1.00	467
	n-gram	**1.20**	**189 (−14%)**	**1.34**	**405 (−13%)**
T4, FT FP16	none	1.00	325	1.00	332
	draft, 2 tokens	1.88	172 (−47%)	1.47	223 (−33%)
	draft, 4 tokens	1.44	210 (−35%)	1.37	240 (−28%)
	draft, 8 tokens	1.33	240 (−26%)	0.96	334 (+1%)
	n-gram, 4 tokens	**1.96**	**162 (−50%)**	**2.44**	**133 (−60%)**

Table 2: Speculative decoding for the fine-tune, one request at a time. CPU speed-up is decode tokens per second relative to plain decoding; T4 speed-up is wall time of plain decoding over wall time of the setting. The draft is Qwen2.5-Coder-0.5B-Instruct (Q4_K_M on the CPU, FP16 on the T4). N-gram lookup proposes continuations found in the prompt and the text so far.

which raises time to first token on repair from 5.7 s to 7.9 s. N-gram lookup proposes tokens at almost no cost, so it wins with lower acceptance (76% and 58%): 1.5× faster, 21–25% less energy, and no extra memory (1.89 GB peak vs. 2.48–2.66 GB with a draft). A Q8 draft was slower than a Q4 draft at equal length, in line with draft latency dominating. With a Q4 target, n-gram lookup saves about half as much (13–14%), since a cheaper target leaves less to save per token; Q4 with n-gram lookup is the cheapest configuration we measured, at 189 J per solved HumanEval+ problem and 405 J per fixed bug.

On the **T4** verification of a block costs little more than one token, so both methods help; short drafts work best (2 tokens), and n-gram lookup is again the best setting, halving energy per correct answer on HumanEval+ and cutting it by 60% on repair, where the answer mostly copies the buggy code in the prompt.

Correctness. Greedy speculative decoding should reproduce plain decoding. Pass counts indeed move by at most two problems in every setting ($p \geq 0.5$), but answers are not bit-identical: on the CPU 14 of 163 HumanEval+ answers differ with the draft model and 8 with n-gram lookup; on the T4 155–164 of 163/164 match. Scoring several tokens in one batch changes floating-point rounding; on the T4 plain decoding at 1 and 8 concurrent requests differs on as many answers.

5 Fine-tuning shifts the target away from its draft

The draft model is the base 0.5B model, so it imitates the base 1.5B model better than the fine-tune. Figure 2 and Table 3 show the cost. Acceptance is lower for the fine-tuned target at every length on both benchmarks, and the gap widens with length and on the task the fine-tune was trained for: on repair it is 0.09 at 2 tokens and 0.24 at 8. With 8 drafted tokens the fine-tune repairs bugs 4% more slowly than without speculation, while the base model still gains 1.47×. For a fine-tuned deployment, a stock draft model should be used with short drafts, or be fine-tuned or distilled together with the target; n-gram lookup has no such mismatch.

	HumanEval+			HumanEvalFix		
Drafted tokens	2	4	8	2	4	8
Acceptance, base target	0.95	0.91	0.84	0.96	0.93	0.87
Acceptance, fine-tuned target	0.94	0.89	0.79	0.87	0.79	0.63
Speed-up, base target	1.60	1.42	1.24	1.40	1.43	1.47
Speed-up, fine-tuned target	1.88	1.44	1.33	1.47	1.37	0.96

Table 3: Draft acceptance on the T4, one request at a time, for the same stock 0.5B draft.

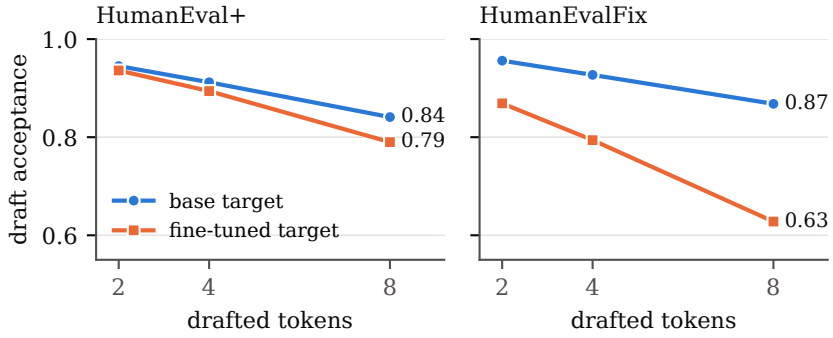


Figure 2: Acceptance of the same stock 0.5B draft for the base and the fine-tuned 1.5B target on the T4, one request at a time.

6 Batching

On the T4, 8 concurrent requests instead of 1 lower energy per correct answer from 325 J to 36 J on HumanEval+ for the fine-tune (9×), and from 1007 J to 142 J on repair for the base model (7×); accuracy is unchanged. This is larger than any speculative setting, and it shrinks speculation's gain: at 8 requests the best draft setting saves at most 19% and some lose energy, while n-gram lookup still saves 14–46%. A local assistant serves one user, so batching is rarely available on a laptop; speculation is the lever that remains. On the same T4, CUDA graphs made a single request 1.6× faster than eager mode, and prefix caching had no measurable effect on repair prompts that share only a short instruction.

7 Do the fine-tuning gains replicate?

We repeated the fine-tune with a second seed (1234 instead of 3407) and the same data, recipe and library versions. Against the base model, both seeds gain on MBPP+ (234 vs. 212 solved for both seeds, $p = 0.005$) and on HumanEvalFix (59 and 58 vs. 41, $p = 0.006$ and 0.008), and neither changes HumanEval+, CRUXEval-O or completion significantly. One result did not replicate: bug detection in HumanEvalPack's yes/no format, where the second seed labels almost every function as buggy (recall 0.95, precision 0.53; accuracy 54.6% vs. 65.6%), although both seeds score 66% in the format they were trained on. We therefore report the generation and repair gains and not the detection gain.

	3B base	A	B	C	D	E
HumanEvalFix (164)	50.0	43.9	**64.6**	61.0	61.6	63.4
reworded prompt	40.2	38.4	**56.7**	–	–	–
3 attempts, explained errors	54.9	52.4	**71.3**	–	–	–
Refactoring, behaviour kept	89.0	**92.0**	78.5	88.3	87.7	87.7
Test generation, bug caught	62.2	**85.4**	82.3	80.5	82.9	82.3
HumanEval+ (163)	**81.0**	74.8	74.8	73.0	72.4	74.8
HumanEvalExplain (164)	**68.3**	–	–	–	57.9	**68.3**

Table 4: Qwen2.5-Coder-3B, % on the T4 (FP16). A: v2 data; B: + realistic bugs; C: B with half the mutations and twice the replay; D: C with corrected explanations; E: D's data with self-distilled replay (deployed). Vs. base on HumanEvalFix: B 35 gained / 11 lost ($p = 5 \cdot 10^{-4}$), D 30 / 11 ($p = 0.004$), E 32 / 10 ($p = 0.0009$); with a reworded prompt that no training example uses, B 41 / 14 ($p = 4 \cdot 10^{-4}$). For reference, Qwen2.5-Coder-7B (4-bit AWQ) fixes 59.2% and the 9B teacher 65.8%.

8 What the fine-tuning data teaches, and a larger base

The 1.5B fine-tune repairs bugs better than its base, but a breakdown by HumanEvalFix bug type shows how: it fixes more operator bugs than the 9B teacher (16 vs. 13 of 25) and barely moves on missing logic ($4 \rightarrow 7$ of 33; teacher: 22). Its training bugs were all one-line mutations, and it learned that shape. The untrained Qwen2.5-Coder-3B already repairs more than our fine-tuned 1.5B (50.0 vs. 36.0%, $p = 0.001$), so we moved to the 3B and changed the data: the teacher wrote one bug of a given kind (missing or excess logic, wrong condition, off-by-one, wrong variable, value or update) into 4,500 unused, verified problems, and a bug was kept only if the problem's own unit tests failed on it, it kept all signatures, and missing/excess-logic bugs removed/added a statement (3,271 kept, 44% spanning several lines). Five QLoRA runs used the same recipe; D retrains C with corrected explanation text, and E replaces 2,648 of C's general replay answers with the base model's own answers to the same prompts:

Table 4 separates the effects. The v2 data alone (A) teaches tests and refactoring but not repair. The realistic bugs (B) add 15 points of repair, the largest gains on excess logic ($12 \rightarrow 21$ of 31) and missing logic ($12 \rightarrow 15$), and the gain survives a reworded prompt, so it is not the benchmark's phrasing. Given a docstring instead of failing tests, the base already fixes 73.8% and B adds nothing (75.0%): what the data teaches is repair *from failing tests*, the situation of a repair agent. Run C restores refactoring at no significant cost in repair (61.0 vs. 64.6%, $p = 0.29$). HumanEval+ stays 6–9 points below the base in every run, including A, so the bug data is not its cause; we suspect the bare-code answers (88 vs. 242 output tokens). Over all eleven tasks, with Holm's correction applied to the nine that were not the training's aim, neither D nor E is significantly below the base; E is the run that loses least (HumanEval+ 74.8%, and it explains code as well as the base, where D drops 10 points, D vs. E $p = 0.019$). We deploy E. It was chosen after these results were known, so its scores are slightly optimistic.

A scoring correction. The benchmark prompt asks for the answer "in a markdown code block", and the base 3B sometimes labels its whole answer `markdown` instead of `python`. Our extraction first skipped such blocks; it now counts one that is valid Python. Six of the base's seven HumanEvalFix answers written this way are correct fixes, which raises the base from 46.3% to 50.0% (refactoring from 86.5% to 89.0%). The fine-tuned models never answer this way, so their scores do not change. Every number in this paper uses the corrected scoring.

For the energy question the length matters too: on the laptop, the fine-tuned 3B answers a repair in 65 tokens against the base 3B's 123, and fixes a bug in about 15 s (Q4_K_M, 2.2 GB RAM) against 21 s for the base and 6 s for the fine-tuned 1.5B. With n-gram lookup on all 164 problems it fixes 100 (61.0%, the same as in FP16 on the T4) for 446 J of CPU energy per fixed bug, against 405 J for the fine-tuned 1.5B in Q4 with n-gram lookup, which fixes 49: twice the repairs for 10% more energy per repair, within the day-to-day drift we measured. Per correct answer, the larger model is not more expensive. Run D repeats the laptop measurement a week later (100 fixed, 456 J per fixed bug) and the deployed run E fixes 95 (478 J per fixed bug; D vs. E 12 / 7 discordant, $p = 0.36$), at the same 15.2 s per problem and 2.2 GB.

9 A pre-registered check of the two claims

The realistic-bug kinds were chosen after reading HumanEvalFix's results by bug type, so the repair gain may be the training distribution meeting a benchmark that resembles it. Before running anything, we committed a protocol fixing the benchmarks, the primary model (run D, then deployed), the prompt, the test and the sentence to report for each outcome.

Repair on real bugs. ConDefects [12] collects faulty and accepted AtCoder submissions; we kept the 382 Python programs from contests held from February 2024, after the base model's data cut-off, whose accepted version passes every test and whose faulty version fails one. Each prompt gives the faulty program and one failing input with its expected and actual output; a program counts as fixed when it passes every test. Run D fixes 57 against the base's 54 (+0.8 points, 95% CI −3.2 to +4.7, $p = 0.79$): as registered, *on real bugs outside the training distribution the repair gain is not confirmed*. Runs C and E behave alike (58 and 56), and the fine-tuned 1.5B is not better than its base (23 vs. 29). On QuixBugs [6] (40 programs, public since 2017 and registered as secondary) the 3B runs gain 17–20 points, which does not pass Holm's correction. These counts use the corrected extraction (Section 8): as first scored, the base fixed 47 ConDefects programs and the difference was +2.6 points ($p = 0.24$); 7 of its 31 `markdown`-labelled answers are correct fixes. The registered outcome is the same under both.

Tests with more room. At 768 output tokens the base might lose only because its test files are cut off. With 2,048 tokens the fine-tune's tests still catch the bug more often (82.9 vs. 60.4%, $p = 10^{-6}$), so the gain holds; but the base is still cut off in 97 of 164 answers, often repeating itself, and where it does finish its tests catch the bug slightly more often (93 vs. 86%). The fine-tune's gain is mostly that it writes short, complete files.

10 Related work

Speculative decoding [1, 5] drafts tokens with a cheap model and verifies them with the target in one pass; Yan et al. [13] show that the speed-up depends mainly on the draft's latency rather than its accuracy, which our CPU results illustrate at the extreme where the draft shares the target's compute. Prompt lookup decoding [11] replaces the draft model with n-gram matches from the context and suits tasks that copy their input, such as repair. Energy measurements of LLM inference have mostly targeted large models on data-centre GPUs [8, 10]; we measure a small model on the hardware it is meant for, and normalise by correct answers.

11 Limitations

One model family, two sizes, two devices, Python only, and one run per configuration except for the seed check; differences under about 10% in time or energy should not be read as findings. The 9B teacher wrote the 3B's training bugs, so the 3B approaching the teacher on HumanEvalFix is a student nearing its teacher on the teacher's kind of bugs, not evidence of general repair skill; the pre-registered check is the stronger test of that. RAPL covers the CPU package and not memory or the rest of the laptop; T4 board energy includes idle draw and so favours whatever finishes sooner. The CPU numbers come from unattended sessions on 14 and 19 October; plain decoding measured on both gave 250 and 244 J per HumanEval+ problem for Q8 and 141 and 131 J for Q4, so cross-session comparisons carry a few percent of drift, and the speculative comparisons are all within one session. Kaggle GPUs are shared.

Reproducibility and use of AI tools

Code, prompts, every model answer, logs and power traces are at `https://github.com/MedGm/Locali`; the merged model and the training data are on Hugging Face. The author used an AI coding assistant (Claude Code) to write much of the evaluation code and experiment scripts and to draft this text, under the author's direction and review; the research questions, design and conclusions are the author's, and every number comes from a committed run.

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